Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Inertia — ANSWERS

Higher tier

1 · Hard to start, hard to stop

Key concept — inertia

Inertia is the tendency of an object to stay at rest, or to keep moving at a steady speed in a straight line.

Everything has inertia. The more mass an object has, the harder it is to change its velocity: harder to start, harder to stop, harder to turn.

Inertial mass is a measure of how difficult it is to change the velocity of an object. It is the resultant force divided by the acceleration it produces.

$$m = F \div a$$

m → **inertial mass** (kg)

a → **acceleration** (m/s^2)

F → **resultant force** (N)

This is $F = ma$ rearranged. An object with a big inertial mass needs a big force to give it even a small acceleration.

Worked example — copy this shape

A trolley. A resultant force of 4 N gives a trolley an acceleration of 0.5 m/s^2. Calculate its inertial mass.

Equation $m = F \div a$

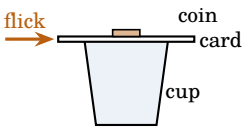
Substitute $m = 4 \div 0.5$

Calculate $m =$ **8 kg**

Check it. Multiply back: $8 \times 0.5 = 4$ N, the force you started with.

2 · The coin and the card

(a) Your teacher balances a card on top of a cup and puts a coin in the middle of the card. Then the card is flicked away sideways, fast.



Describe what happens to the coin. Explain why, using the word **inertia**.

The coin stays where it was, then drops straight down into the cup. The flick pushes the card, not the coin. The only force on the coin from the card is a small friction force for a very short time, so the coin's velocity hardly changes: its inertia keeps it at rest. Once the card has gone, its weight is unbalanced and it falls into the cup.

[4]

(b) An empty shopping trolley and a full one are pushed with the same resultant force. Which one has the bigger acceleration? Explain your answer using **inertial mass**.

The empty trolley. It has the smaller inertial mass, so the same force changes its velocity more easily: $a = F \div m$ is bigger.

[2]

(c) (i) A resultant force of 12 N gives a trolley an acceleration of 0.8 m/s^2. Calculate its inertial mass.

$m = F \div a = 12 \div 0.8 =$ **15 kg**.

(ii) A 3 kg bag is put in the trolley from part (i). The same 12 N resultant force acts on it. Calculate the new acceleration.

New mass $= 15 + 3 = 18$ kg. $a = F \div m = 12 \div 18 =$ **0.67 m/s^2**.

[4]

(d) A bus brakes suddenly. The passengers standing inside lurch forwards. Explain why, using **inertia**.

The brakes slow the bus, not the passengers. Their inertia keeps them moving forwards at the old speed until a force (friction on their feet, a handrail) slows them too.

[2]

(e) **Stretch.** Two crates float side by side inside the International Space Station. One crate has ten times the mass of the other. An astronaut pushes each one with the same force. Which is harder to get moving? Explain your answer.

The crate with more mass. Floating does not remove inertia: inertial mass belongs to the object, wherever it is. Ten times the mass means the same force gives one tenth of the acceleration.

[2]